

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Comparative Analysis

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COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 10%

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

4. Depth Estimation Comparison (Stereo vs SfM)

Estimated depth values (in meters):

Object Stereo Vision Structure from Motion

A	2.0	2.5
B	3.0	3.8
C	4.0	5.0

Actual depths: A=2, B=3, C=4

(a) Compute absolute error for both methods.

(b) Compare and evaluate which method is more accurate.

ANS:

Depth Estimation Comparison — Stereo Vision vs Structure from Motion Given Data

Object	Actual Depth (m)	Stereo Vision (m)	Structure from Motion (m)
A	2.0	2.0	2.5
B	3.0	3.0	3.8

	Object Actual Depth (m)	Stereo Vision (m)	Structure from Motion (m)
C	4.0	4.0	5.0

(a) Compute Absolute Error for Both Methods — 20 Marks

The **absolute error** is calculated as:

Stereo Vision

Object A

Object B

Object C

Therefore:

Object Actual Stereo Estimated Absolute Error

A	2.0	2.0	0.0 m
B	3.0	3.0	0.0 m
C	4.0	4.0	0.0 m

Total and Mean Absolute Error

Structure from Motion

Object A

Object B

Object C

Therefore:

Object Actual SfM Estimated Absolute Error

A	2.0	2.5	0.5 m
B	3.0	3.8	0.8 m
C	4.0	5.0	1.0 m

Mean Absolute Error

(b) Compare and Evaluate — 15 Marks

The comparison is:

Method	Total Absolute Error	Mean Absolute Error
Stereo Vision	0 m	0 m
Structure from Motion	2.3 m	0.767 m

Stereo Vision produces exactly the actual depth values for all three objects.

Structure from Motion consistently **overestimates the depth**:

- Object A → 0.5 m error
- Object B → 0.8 m error
- Object C → 1.0 m error

Since lower absolute error indicates better accuracy:

Therefore:

Evaluation

Stereo Vision is preferable for this particular dataset because it has **zero error**. It provides more precise depth measurements than Structure from Motion. However, in real-world applications, the performance of stereo vision depends on factors such as camera calibration, baseline, image quality, and correspondence matching. SfM can be useful when multiple images from different viewpoints are available and a stereo camera setup is not available.

5. Pattern Recognition Accuracy Comparison

Two recognition systems produce results:

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

(a) Compute accuracy for both systems.

(b) Compare and evaluate which system performs better.

ANS:

Pattern Recognition Accuracy Comparison

30 Marks

Given Data

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

(a) Compute Accuracy — 15 Marks

The accuracy formula is:

Traditional System

Deep Learning System

Therefore:

System	Correct Predictions	Total Samples	Accuracy
Traditional	85	100	85%
Deep Learning	92	100	92%

(b) Compare and Evaluate — 15 Marks

The difference in accuracy is:

Therefore, the Deep Learning system has a **7 percentage-point higher accuracy**.

Number of incorrect predictions:

Traditional

So, Traditional system makes **15 incorrect predictions**.

Deep Learning

So, Deep Learning makes only **8 incorrect predictions**.

Thus:

System	Accuracy	Incorrect Predictions
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Traditional	85%	15
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Deep Learning	92%	8
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Evaluation

The Deep Learning system performs better because it correctly recognizes **92 out of 100 samples**, compared with only **85 out of 100** for the Traditional system.

Deep learning can learn complex patterns and features automatically, whereas traditional pattern-recognition techniques may depend more heavily on manually selected features.

The improvement is:

Therefore:

6. Precision-Recall Trade-off Analysis

Two models produce:

Model	TP	FP	FN
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A	70	30	20
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B	85	25	25
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(a) Compute precision and recall for both models.

(b) Compare and evaluate which model is more balanced.

ANS:

Precision–Recall Trade-off Analysis

35 Marks

Given Data

Model	TP	FP	FN
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A	70	30	20
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B	85	25	25
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(a) Compute Precision and Recall — 20 Marks

Precision

Precision measures how many of the samples predicted as positive are actually positive.

Recall

Recall measures how many of the actual positive samples are correctly detected.

Model A

Given:

Precision of Model A

Recall of Model A

Model B

Given:

Precision of Model B

Recall of Model B

Results

Model	TP	FP	FN	Precision	Recall
A	70	30	20	70%	77.78%
B	85	25	25	77.27%	77.27%

(b) Compare and Evaluate Which Model Is More Balanced — 15 Marks

A balanced model should have **both high precision and high recall**, rather than performing well in only one measure.

Model A

Difference:

Model A has better recall than precision, meaning it detects a relatively high proportion of actual positive cases but produces more false positives.

Model B

Difference:

Therefore, Model B has almost perfectly balanced precision and recall.

F1-Score for Further Evaluation

The F1-score combines precision and recall:

Model A

Model B

Since precision and recall are both approximately 77.27%:

Therefore:

Model	Precision	Recall	F1-Score
A	70%	77.78%	73.68%
B	77.27%	77.27%	77.27%

Evaluation

Model B is more balanced because:

- It has higher precision than Model A.
- It has almost the same recall as Model A.
- Precision and recall are exactly balanced.
- It has a higher F1-score.
- It has fewer false positives than Model A.

Model A has a slight advantage in recall:

but the difference is only:

Model B has a much better precision:

Therefore, the overall performance favors Model B.